New Software Development Processes for Interplanetary Small Satellites

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Avionics Software is core to spaceflight

- Software governs spacecraft behavior
- Large Percentage of Development
- Small spacecraft must do many things larger missions do
- In some ways, its harder
  - COTS, Lower Costs Components

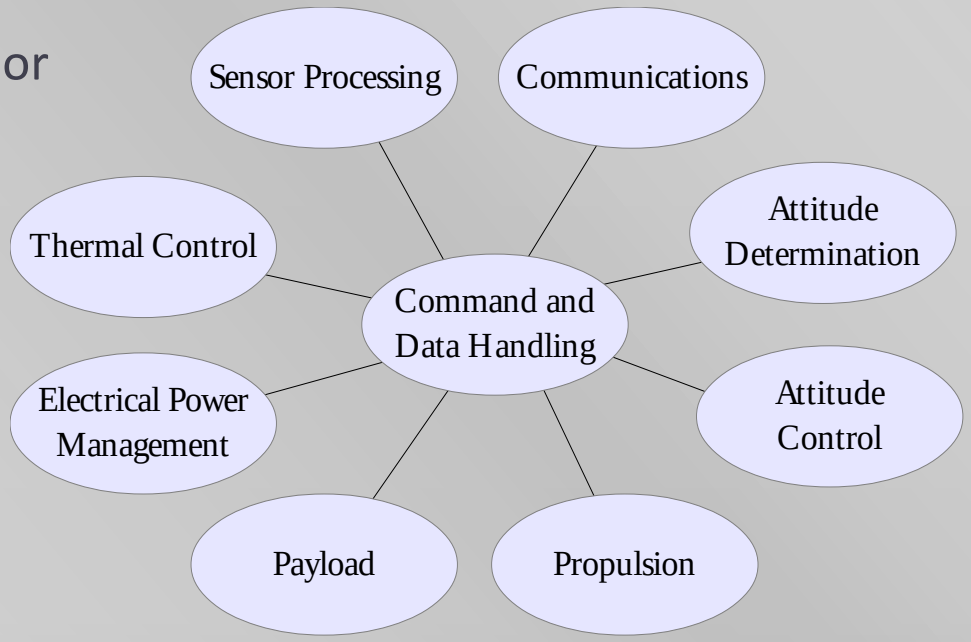
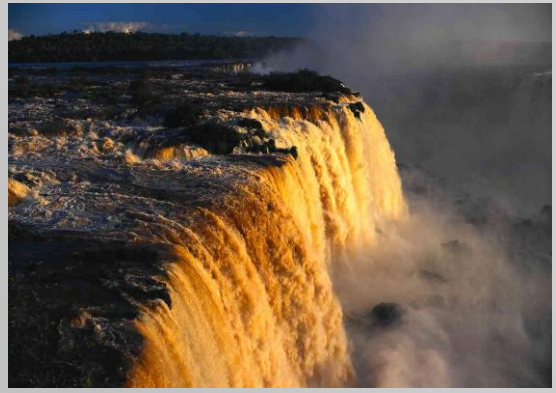


Figure 1: Subsystem Context Diagram Showing Common Spacecraft Flight Systems

Existing Development Strategies



- Sequential Development Common:
  - Waterfall
- Reviews between phases
- Each phase complete before moving to the next
- Designed for Large Teams
- Overhead

Requirements → Preliminary Design → Critical Design → Code Implementation → Integration → Test

Agile and Iterative Methods

- Popular Modern Software Development Practices
- Developed to provide lean methods for engineering software systems
- Focus on Collaboration
- Focus on developing working software quickly, while minimizing focus on process
- Changes in development scope encouraged

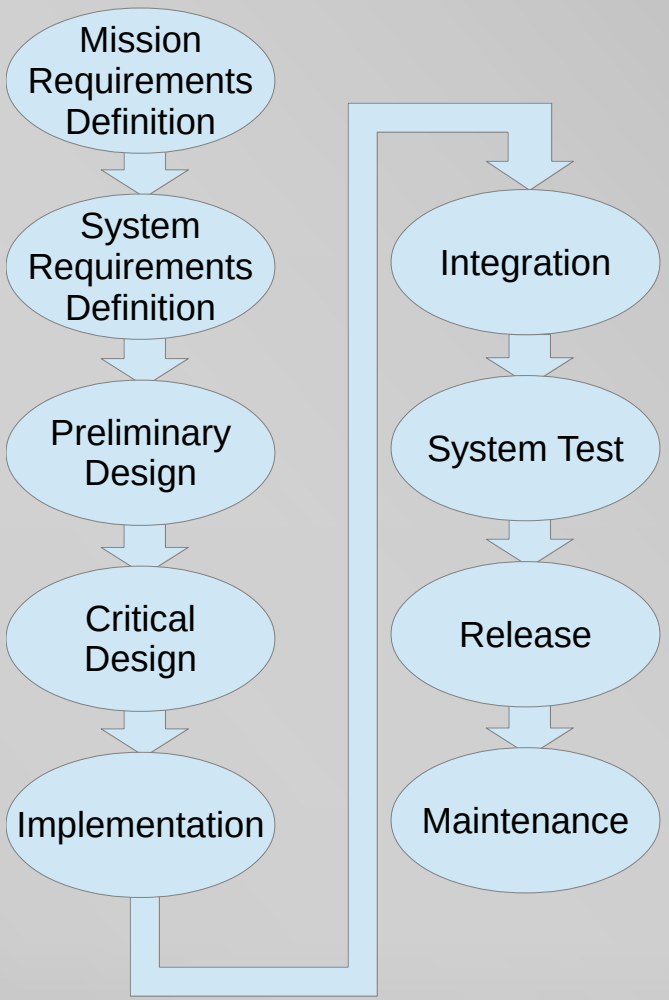


Figure 2: Example Waterfall Project Flow

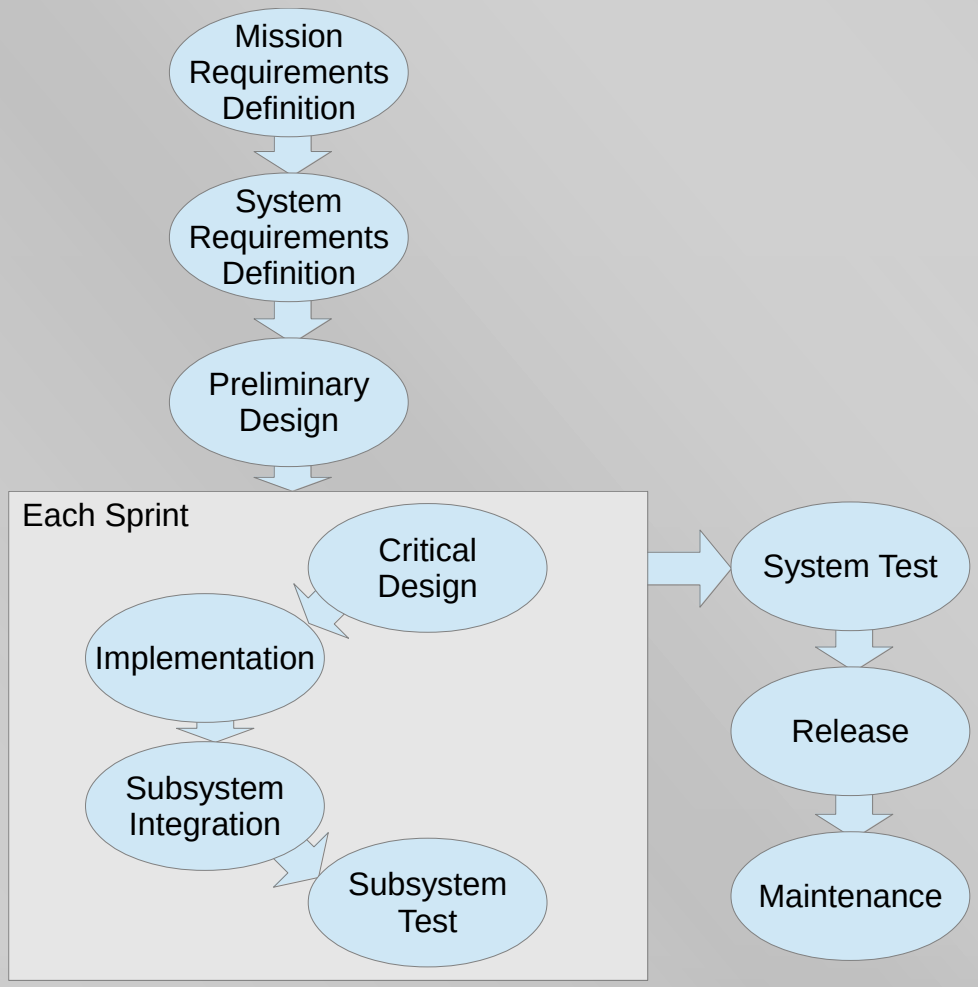
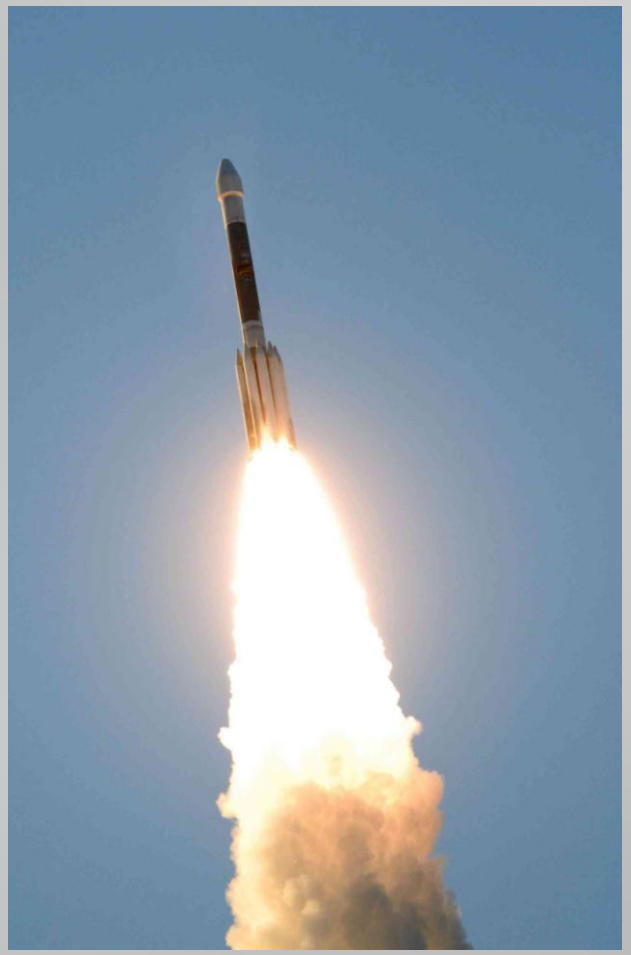


Figure 3: Example Agile Project Flow

Importance of Development Methodology

- Constrained Schedules and Budgets
- Available launch opportunities can drive schedule
- Numerous studies show Agile providing greater than 40% productivity improvements over Waterfall methods
- Adaptation



Workflow Oriented Balance

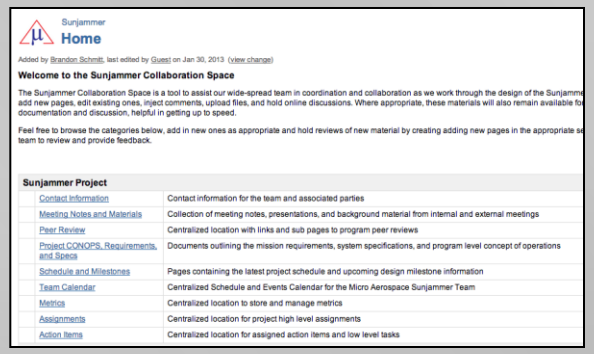
- Not just Copy-Paste
- Spaceflight Reliability, Maintainability, Documentation Constraints
- Rapid Product Development vs Process
- Workflow Oriented Approach
- Tools to support documentation and process requirements

Case Study: Sunjammer

- Technology Demonstration Mission
- Demonstration of a deployable Solar Sail
- Demonstration of propellant-less interplanetary navigation
- Approx 2.5 years from Systems Requirements Review to Launch
- Low cost mission

Case Study: Sunjammer

- Applying Agile Development Methodologies
- Requirements Definition, Preliminary Design
- Critical Design and Implementation Phase
- From Agile:
  - 2 Week Time Boxed Sprints
  - Regular team sync ups
  - Sprint scope plan can change prior to the Sprint
- To Support Process
  - NASA 7150.2a and 7120.5e compliant processes
  - Documentation, peer reviews, design artifacts tools



Sunjammer
Home

Added by Brandon Schmitt, last edited by Guest on Jan 30, 2013 (view change)

Welcome to the Sunjammer Collaboration Space

The Sunjammer Collaboration Space is a tool to assist our wide-spread team in coordination and collaboration as we work through the design of the Sunjamme add new pages, edit existing ones, inject comments, upload files, and hold online discussions. Where appropriate, these materials will also remain available fo documentation and discussion, helpful in getting up to speed.

Feel free to browse the categories below, add in new ones as appropriate and hold reviews of new material by creating adding new pages in the appropriate se team to review and provide feedback.

Sunjammer Project	
Contact Information	Contact information for the team and associated parties
Meeting Notes and Materials	Collection of meeting notes, presentations, and background material from internal and external meetings
Peer Review	Centralized location with links and sub pages to program peer reviews
Project CONOPS, Requirements, and Specs	Documents outlining the mission requirements, system specifications, and program level concept of operations
Schedule and Milestones	Pages containing the latest project schedule and upcoming design milestone information
Team Calendar	Centralized Schedule and Events Calendar for the Micro Aerospace Sunjammer Team
Metrics	Centralized location to store and manage metrics
Assignments	Centralized location for project high level assignments
Action Items	Centralized location for assigned action items and low level tasks

Summary

- Avionics software is a critical component
- Agile Development Methods can help improve productivity and adaptability to changes in scope
- The use of tools to support process and reliability needs
- These methods are being applied successfully on Sunjammer, a small interplanetary spacecraft development program

Questions